SignBridge

A signing companion that shows its gaps.

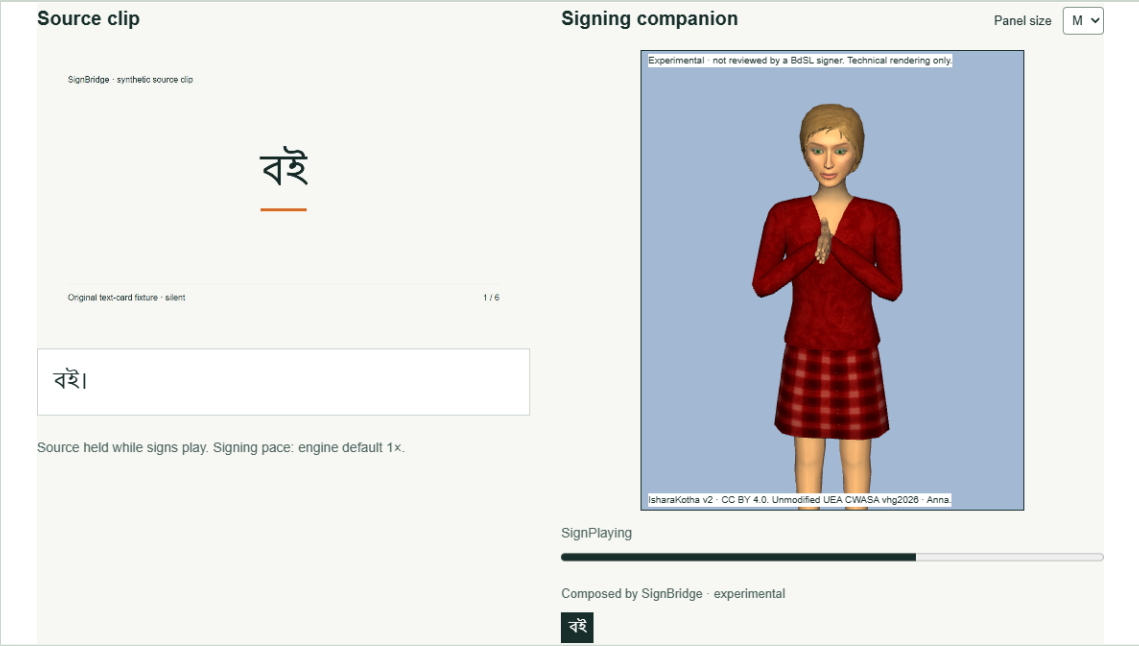
- Design target: Deaf and hard-of-hearing viewers and Bangla video creators. This audience hypothesis has not been validated with participants.

- Short captioned clips become saved, source-timed experimental signing tracks. Missing critical concepts stay visible before signing.

- Every new composition is labelled Experimental - not reviewed by a BdSL signer.

SYSTEM-DESIGN.md section 0.3; phase reports 2, 5 and 7. Original synthetic demo content; no user research claimed.

Follow the source, then inspect the gap



Covered production segment: source held while the real avatar signs



Unsupported production negation: blocking gap before signing

Upload captions -> validate a plan -> watch the source -> hold for signs -> replay the saved track.

Production sample A: 45 seconds, six technically ready segments, 12 model calls. Sample B deliberately leaves negation unsupported. No plan was hand-edited.

Publish a durable companion track



Neon SQL: owner, epoch, stage cache, budgets, versioned manifest

Private Blob: signed source video | UEA CWASA: isolated signing iframe

- Propose meaning and corpus-asset steps; validate structure and critical gaps; allow one repair, then persist the result.

- Source-caption edits stop affected signing. SQL authority and epochs fence stale publishers; disposable renderer instances reject old callbacks.

- Source and avatar use separate clocks. Exact ordered unit completion counts; timeout, missing callback and skip do not count as completed signing.

phase reports 3-7; T6/T7/T9/T16/T20/T22. One actual Workflow survived production cutover and published exactly one track.

Keep three evidence categories separate

Category	Current evidence
Technical renderability	20/20 sampled corpus files in Phase 2; 12/12 curated assets in Phase 7. One Windows host, software WebGL. Six production segments replayed in Chrome and Edge.
Linguistic correctness	None. No qualified BdSL signer assessed these exact rendered tracks.
User comprehension	None. No evaluation using tasks with Deaf/BdSL users.

Engineering checks: 194 release tests, 51 native PostgreSQL tests, 16 local browser tests; production T1/T2/T9/T14 pass. These are not language-quality scores.

SignOLight documents video-to-avatar; IsharaKotha supplies prior text-to-avatar research and corpus assets. Our tested contribution is the combination of source-frame holds, durable caption edits, saved replay and visible critical gaps. No firstness claim.

phase reports 2-7 and dated phase-6/7 rechecks; SYSTEM-DESIGN.md H.3; competition-event.md prior-art audit (3 November 2026).

Solo build, followed by formative evaluation

- Built as a solo project using AI coding tools. Next: consented formative sessions with qualified Deaf/BdSL reviewers, assessing exact rendered versions and critical gaps.

- Run separate comprehension tasks with Deaf/BdSL users; report participant numbers, errors and uncertainty before making outcome claims.

- Check real Android and iPhone performance. The owner supplied the official event sticker; it is used unchanged in the top banner.

Try the two public, read-only synthetic samples from the production home page.

phase-7 report; docs/MOBILE-TEST.md; SYSTEM-DESIGN.md H.3/H.6. Linguistic and comprehension evaluation are planned, not performed.